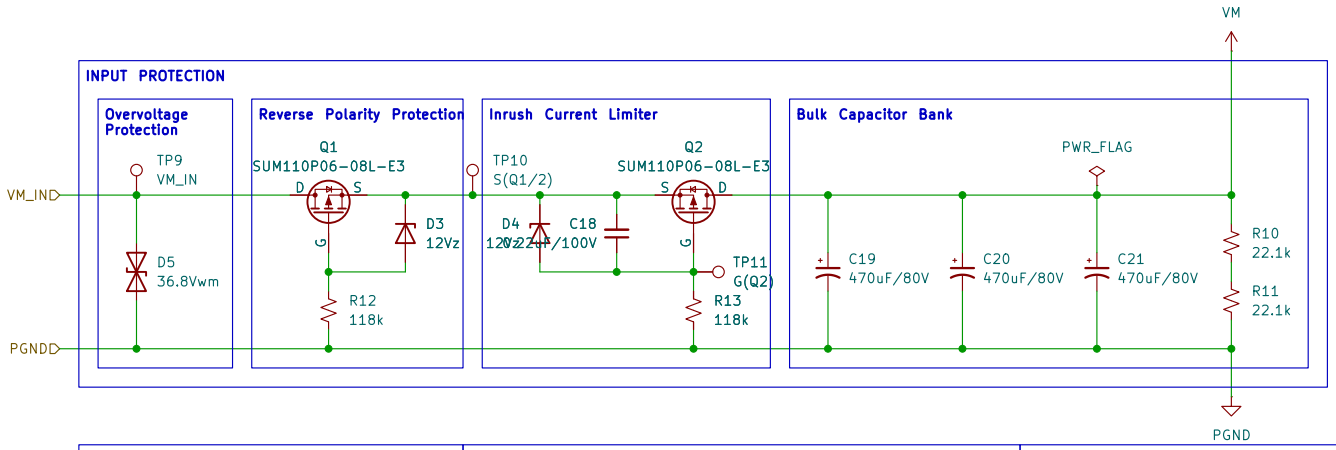


Absolute Maximum Ratings
- Supply Voltage: 36V. Continuous Current: 15A. Worst Load: 36V, 250W E-bike Motor.

Bulk Capacitor Bank
- C_bulk(min) = [I(max)*D(1-D)]/(f_sw*deltaV_ripple(target)) = (15*0.25)/(20000*0.36) = 3.75/7200 = 520.8uF
- I_ripple(total) > I(max)*sqrt[D(1-D)] = 15*sqrt(0.25) = 7.5A_rms => I_ripple(single) > 7.5/3 = 2.5A_rms

- C_total = 3*470uF = 1410uF. ESR_total = 30mOhms/3 = 10.0mOhms (in parallel, 100kHz/20°C)
- coefficient_log-interpolated(20kHz) = 0.986 (Panasonic factor table, 390-1000uF/6.3-100V bracket) => I_ripple(single) = 2.2*0.986 = 2.17A_rms @ 20kHz => I_ripple(total) = 3*2.17 = 6.51A_rms @ 20kHz, below the 2.5A/7.5A floor above; acceptable per 100% non-continuous 15A loading
- E = 0.5*C*V^2 = 0.5*1410uF*36^2 = 0.914 J

- I_bleed = 36V / 44.2kOhms = 814.5uA => P_bleed(single) = 18V*814.5uA = 14.66mW
- tau_bleed = R_total x C_total = 44.2kOhms× 1290uF = 62.3s

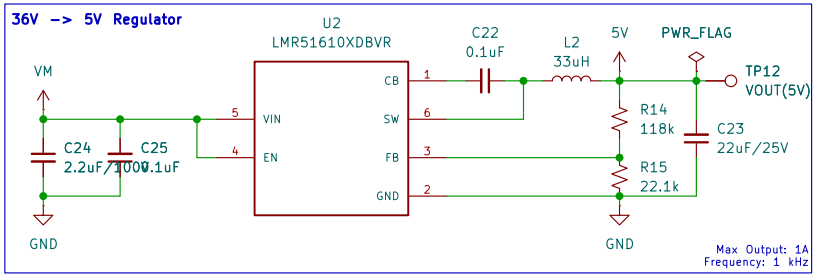
MOSFETs
- T_a=70°C, R_th(J-A)=40°C/W: R_DS(on)=9.6mOhms (graph, ~150°C), P=2.16W, T_J=156.4°C => 18.6°C to 175°C(max)

TVS Diode OVP
- PMOS has V_DS(max)=60V, so V_C=59.3V<60V
- V_spec(max)=36V, so V_wm=36.8V>36V

High-side PMOS RPP
- PMOS has R_DS(on) = 5.5mOhms typ / 6.8mOhms max @ -10V and V_GS(max)=+/-20V, so V_Z=12V
- tau = R_pd*C_iss = 118kOhms*10200pF = 1.20ms << 27.16ms

RC Soft-Start
tau(typ) = R_pd*[C_ss+C_iss] = 118kOhms*(220nF+10.2nF) = 27.16ms > 1.20ms

Power on PMOS' Zener Diode and Pull-down Resistor
- P_Z = V_Z*I_pd = 12V * 203.4uA = 2.44mW < 500mW
- V_G = V_bus-V_Z = 36-12 = 24V. I_pd = 24V/118kOhms = 203.4uA.
- P_pd = V_G*I_pd = 24V*203.4 uA = 4.88mW < 0.1W



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